Gov 51 Pset 4

Caleb Masshardt

GitHub Repository: https://github.com/cmasshardt/gov51-ps4.git

1. Setup and Exploration

1.1. Load packages and data

```
#Reading in libraries
library(tidyverse)
library(estimatr)
library(modelsummary)

#Reading in card1995 dataset
card <- read_csv("./data/card1995.csv")
```

1.2. Describe the data

How many observations are in the dataset? What is the unit of observation?

```
nrow(card)
```

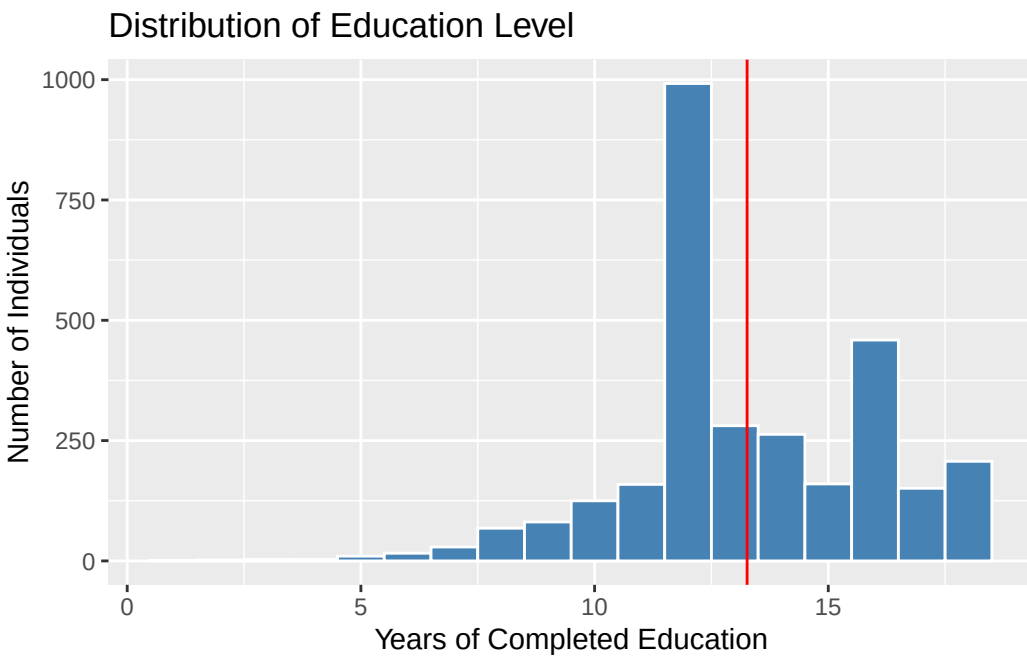
```
[1] 3010
```

There are 3,010 observations in the data set. The unit of observation is an individual participant in the survey — a young man of age 14-24 as of 1966.

1.3. Distribution of years of education

Create a histogram of educ. What is the mean? Are there any unusual spikes? What might explain them?

```r
#Plotting histogram
ggplot(card, aes(x = educ)) +
  geom_histogram(binwidth = 1,
                 fill = "steelblue",
                 color = "white") +
  geom_vline(xintercept = mean(card$educ), color = "red") +
  labs(x = "Years of Completed Education",
       y = "Number of Individuals",
       title = "Distribution of Education Level")
```




```r
#Calculating mean
mean(card$educ)
```

```
[1] 13.26346
```

The mean value for educ (years of completed education) is 13.26346.

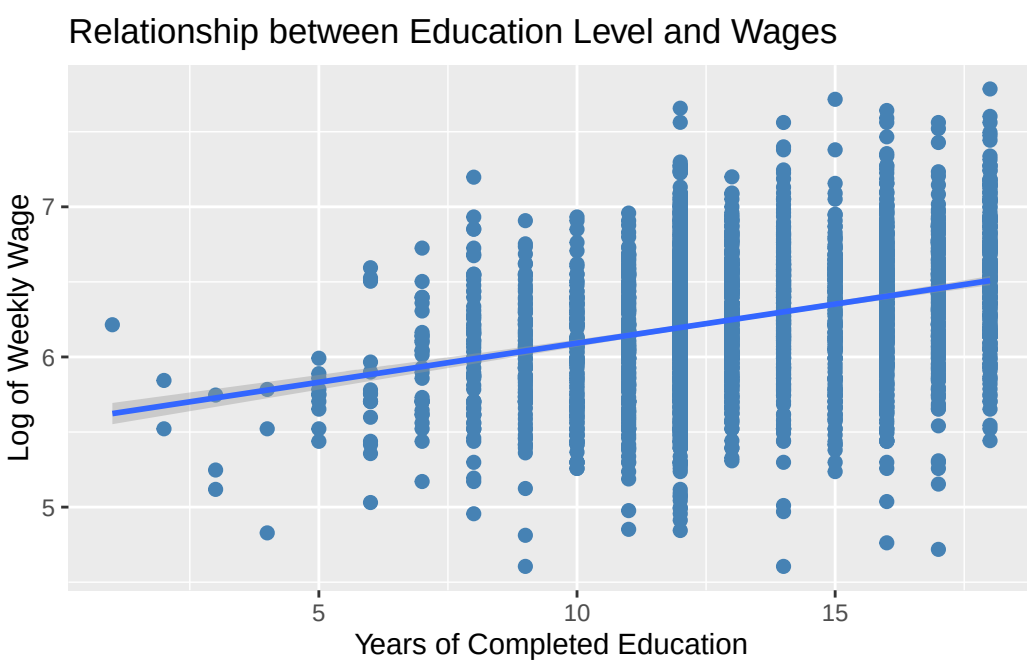
The two prominent spikes in the histogram are at 12 years and 16 years, which make sense given that high school (12 years) and bachelor's degree (12 + 4 = 16 years) are the most common education levels and serve as cutoffs for levels of education.

1.4. The naïve relationship

Create a scatter plot of lwage (y-axis) against educ (x-axis). Add a linear regression line. Based on this plot, does more education appear to be associated with higher wages?

```
ggplot(card,
       aes(x = educ, y = lwage)) +
  geom_point(color = "steelblue", size = 2) +
  geom_smooth(method = "lm") +
    labs(x = "Years of Completed Education",
         y = "Log of Weekly Wage",
         title = "Relationship between Education Level and Wages")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Yes – based on this scatterplot and line of best fit, there is clearly a positive association between level of completed education and wages. As such, more education *appears* to be associated with higher wages.

1.5. Why OLS might be misleading

The OLS regression of lwage on educ might not give us the causal effect of education. Name two specific reasons why people with more education might earn more even if education itself had no effect.

One potential reason is that socioeconomic class (family background) influences both education level and earnings — those in higher income brackets are more likely to go to college *and* are more likely to earn higher wages due to family connections, cultural similarity with those at elite, high-paying firms, and other class-related advantages. Another reason could be the confounder of talent and work ethic (ability bias) — those with the skills and motivation to work hard succeed in education, but these especially hard-working types would likely also earn more even without the effect of education because of the value they bring to employers through productivity.

---

2. The Instrument: College Proximity

2.1. What the instrument measures

The variable nearc4 equals 1 if the respondent grew up near a 4-year college. What share of respondents in this sample grew up near a 4-year college?

```
#Since nearc4 is binary, we can use the mean to calculate the proportion
# of respondents who
mean(card$nearc4)
```

```
[1] 0.6820598
```

68.21% of respondents in this sample grew up near a 4-year college.

2.2. Relevance: does proximity affect education?

The first condition for a valid instrument is relevance: the instrument must be correlated with the endogenous variable (education).

Run the following regression — the first stage — and report the coefficient on nearc4 along with its standard error and t-statistic:

```
first_stage <- lm_robust(educ ~ nearc4 + black + smsa + south +
exper + expersq + married,
data = card, se_type = "HC2")

fs_summary <- summary(first_stage)
fs_summary
```

```
Call:
lm_robust(formula = educ ~ nearc4 + black + smsa + south + exper +
    expersq + married, data = card, se_type = "HC2")

Standard error type:  HC2

Coefficients:
            Estimate Std. Error  t value  Pr(>|t|)  CI Lower  CI Upper   DF
(Intercept) 16.94691   0.160943 105.2976 0.000e+00 16.631341 17.262481 2995
nearc4       0.32711   0.080554   4.0607 5.017e-05  0.169164  0.485059 2995
black       -0.94529   0.088599 -10.6694 4.142e-26 -1.119011 -0.771571 2995
smsa         0.42333   0.085211   4.9680 7.144e-07  0.256251  0.590405 2995
south       -0.30018   0.078571  -3.8204 1.359e-04 -0.454235 -0.146117 2995
exper       -0.43245   0.032388 -13.3522 1.520e-39 -0.495950 -0.368941 2995
expersq      0.00141   0.001707   0.8259 4.089e-01 -0.001937  0.004757 2995
married     -0.07441   0.017670  -4.2109 2.618e-05 -0.109055 -0.039761 2995

Multiple R-squared:  0.4775 ,   Adjusted R-squared:  0.4763
F-statistic: 523.3 on 7 and 2995 DF,  p-value: < 2.2e-16
```

Coefficient on nearc4 = 0.32711

Standard error of coefficient = 0.080554

t-statistic = 4.0607

Interpret the coefficient on nearc4 in a sentence. Does the sign make sense?

Those who live near a four-year college at age 14 have, on average, complete 0.32711 years more of education than those who did not live near a four-year college at age 14.

2.3. Checking instrument strength

Report the t-statistic on nearc4 from the first-stage regression. Compute the first-stage F-statistic as $= t^2$.

```r
#t-statistic on nearc4
t_stats <- coef(fs_summary)[,"t value"]
t_nearc4 <- t_stats["nearc4"]
t_nearc4
```

```
  nearc4
4.060749
```

```r
#Calculating F-statistic
f_nearc4 <- (t_nearc4)^2
f_nearc4
```

```
  nearc4
16.48968
```

The t-statistic on nearc4 from the first-stage regression is 4.0607.

The F-statistic computed as $F = t^2$ equals 16.4897.

Is your instrument strong? (Rule of thumb: $F > 10$ suggests a strong instrument.) Why does instrument strength matter for the reliability of the 2SLS estimate?

Yes, our instrument of nearc4 is strong, since its F-statistic is 16.4897, which is greater than 10. Instrument strength matters for the reliability of the 2SLS estimate because a strong instrument functions to remove/minimize the bias in assignment of the treatment and ensures that there is not too much variation across samples. A weak instrument fails to remove bias compared to OLS and makes the denominator of the 2SLS estimate close to zero, which results in a high variance of estimates across samples.

2.4. Exclusion restriction

The second condition for a valid instrument is the exclusion restriction: college proximity must affect wages only through its effect on education — not through any other channel.

(a) State the exclusion restriction formally: what must be true about the relationship between nearc4 and lwage once we control for education?

The covariance between the instrument Z and the error term is zero. The relationship between the instrumental variable (nearc4) and the outcome (lwage) must run only through the predictor variable (education). The variables nearc4 and lwage must have no covariance once we control for education.

(b) Can you think of a way the exclusion restriction might be violated — that is, a reason why growing up near a college might affect your wages for reasons other than more education? Be specific.

One reason why growing up near a college might affect your wages for reasons other than education is the role of peer network groups. living near a college can expose people to a

higher-status or higher-achieving peer group, which can lead to higher wages outside of the influence on education. In this way, growing up near a college can affect wages for reasons other than education, as the networking benefits of a peer group might operate to increase wages regardless of education level.

2.5. Balance check

If the instrument is as-good-as-randomly assigned, respondents who grew up near a college should look similar on pre-treatment characteristics to those who did not.

Compute the mean black, smsa, and south for the two groups (nearc4 == 0 and nearc4 == 1). Are the groups balanced? What does any imbalance tell you about the instrument?

```
#Subset for those who lived near a college at age 14
card_nearc4 <- card[card$nearc4==1,]

#Subset for those who did not live near a college at age 14
card_not_near <- card[card$nearc4==0,]

#Creating vectors for a summary table
var_names <- c("Percent Black", "Percent Lived in Metropolitan Area (SMSA)",
               "Percent Lived in the South")
nearc4_yes <- c(mean(card_nearc4$black), mean(card_nearc4$smsa),
                mean(card_nearc4$south))
nearc4_no <- c(mean(card_not_near$black), mean(card_not_near$smsa),
               mean(card_not_near$south))

group_means <- data.frame(var_names, nearc4_yes, nearc4_no)

#Table presentation
library(knitr)
group_means |>
kable(
col.names = c("Variable", "Lived Near a 4-Year College",
              "Did Not Live Near a 4-Year College"),
caption = "Comparing Group Characteristics"
)
```

Table 1: Comparing Group Characteristics

Variable	Lived Near a 4-Year College	Did Not Live Near a 4-Year College
Percent Black	0.2118850	0.2800418
Percent Lived in Metropolitan Area (SMSA)	0.8222114	0.4785789
Percent Lived in the South	0.3292742	0.5632184

While the groups are racially balanced for the most part, they are not balanced in terms of the proportion of each that lived in a metropolitan area and that lived in the South. Race is somewhat closer, but still unbalanced. Whereas 82.2% of those who lived near a 4-year college lived in a metropolitan area, only 47.9% of those who did not live near a 4-year college were in a metropolitan area. Moreover, whereas Whereas 32.9% of those who lived near a 4-year college lived in the South, 56.3% of those who did not live near a 4-year college were in a metropolitan area. These geographic differences in the two groups indicate that nearc4 as an instrument is not perfectly balanced. Since the instrument is not independent from other covariates in the data, we need to control for these covariates in our regression.

---

3. The Wald Estimator

3.1. Compute the reduced form effect

Run the reduced form regression: regress lwage directly on nearc4 (without educ, but including the same controls as the first stage). Report the coefficient on nearc4 and interpret it: what does it tell us about how proximity to a college relates to wages?

```
reduced_form <- lm_robust(lwage ~ nearc4 + black + smsa + south +
exper + expersq + married,
data = card, se_type = "HC2")

rf_summary <- summary(reduced_form)
rf_summary
```

```
Call:
lm_robust(formula = lwage ~ nearc4 + black + smsa + south + exper +
    expersq + married, data = card, se_type = "HC2")
```

```
Standard error type:  HC2

Coefficients:
             Estimate Std. Error t value  Pr(>|t|)  CI Lower  CI Upper   DF
(Intercept)  6.104784  0.0369400 165.262 0.000e+00  6.032354  6.177214 2995
nearc4       0.040854  0.0160993   2.538 1.121e-02  0.009287  0.072421 2995
black       -0.236173  0.0183381 -12.879 5.575e-37 -0.272130 -0.200217 2995
smsa         0.196879  0.0166562  11.820 1.527e-31  0.164220  0.229537 2995
south       -0.146503  0.0163740  -8.947 6.243e-19 -0.178609 -0.114398 2995
exper        0.040916  0.0067800   6.035 1.787e-09  0.027622  0.054210 2995
expersq     -0.001789  0.0003228  -5.543 3.240e-08 -0.002422 -0.001156 2995
married     -0.038826  0.0038392 -10.113 1.155e-23 -0.046353 -0.031298 2995

Multiple R-squared:  0.2171 ,   Adjusted R-squared:  0.2152
F-statistic: 128.2 on 7 and 2995 DF,  p-value: < 2.2e-16
```

The coefficient on nearc4 for the reduced form regression is 0.040854. Those who lived near a 4-year college at age 14 on average have a 0.040854 higher log of weekly wages (earn 4.1% more in weekly wages). This tells us that closer proximity to a 4-year college is associated with higher wages.

3.2. Calculate the Wald estimator by hand

Using the first-stage coefficient from Section 2.2 and the reduced-form coefficient from Section 3.1, compute the Wald estimator by hand. Show your calculation.

```r
#Calculating Wald estimator by hand
0.040854/0.32711
```

```
[1] 0.1248938
```

Wald estimator = (reduced form coeff on nearc4)/(first stage coeff on nearc4)

= (0.040854)/(0.32711) = 0.124894

3.3 Interpret the Wald estimator

Interpret your Wald estimate in a sentence. What does it say about the effect of one additional year of education on wages?

The Wald estimate of 0.1249 indicates that each additional year of completed education causes a 12.49% increase in weekly wages.

---

4. OLS vs. 2SLS

4.1. OLS estimate

Run OLS of lwage on educ and the controls. Report the coefficient on educ.

```
ols <- lm_robust(lwage ~ educ + black + smsa + south +
exper + expersq + married,
data = card, se_type = "HC2")

ols
```

```
                Estimate   Std. Error   t value     Pr(>|t|)     CI Lower
(Intercept)  4.902202008 0.0712861999 68.767896 0.000000e+00  4.762427137
educ         0.071498271 0.0036111667 19.799216 4.316524e-82  0.064417653
black       -0.168717410 0.0173999796 -9.696414 6.533265e-22 -0.202834531
smsa         0.172430654 0.0150653394 11.445521 1.017615e-29  0.142891194
south       -0.127691620 0.0151740941 -8.415107 5.996475e-17 -0.157444322
exper        0.071812773 0.0067378450 10.658122 4.654695e-26  0.058601501
expersq     -0.001889027 0.0003149664 -5.997550 2.242922e-09 -0.002506599
married     -0.033550125 0.0035795349 -9.372761 1.348435e-20 -0.040568721
                CI Upper   DF
(Intercept)  5.041976879 2995
educ         0.078578889 2995
black       -0.134600289 2995
smsa         0.201970114 2995
south       -0.097938918 2995
exper        0.085024046 2995
expersq     -0.001271454 2995
married     -0.026531529 2995
```

The coefficient on educ is 0.0715.

4.2. 2SLS estimate

Run the 2SLS regression using iv_robust() from estimatr. The formula syntax separates the structural equation (left of |) from the first-stage instruments (right of |):

```r
iv <- iv_robust(lwage ~ educ + black + smsa + south +
exper + expersq + married |
nearc4 + black + smsa + south +
exper + expersq + married,
data = card, se_type = "HC2")

iv
```

```
               Estimate   Std. Error   t value     Pr(>|t|)     CI Lower
(Intercept)  3.988229477 0.8399956105  4.747917 2.151745e-06  2.341202728
educ         0.124893231 0.0490790159  2.544738 1.098575e-02  0.028661238
black       -0.118113004 0.0493318422 -2.394255 1.671531e-02 -0.214840728
smsa         0.144007839 0.0305020895  4.721245 2.451737e-06  0.084200673
south       -0.109013471 0.0230566857 -4.728063 2.371435e-06 -0.154222014
exper        0.094925153 0.0223207958  4.252767 2.176073e-05  0.051159510
expersq     -0.001965169 0.0003465968 -5.669900 1.564817e-08 -0.002644761
married     -0.029532543 0.0051528211 -5.731335 1.095519e-08 -0.039635970
               CI Upper   DF
(Intercept)  5.635256227 2995
educ         0.221125225 2995
black       -0.021385280 2995
smsa         0.203815006 2995
south       -0.063804928 2995
exper        0.138690796 2995
expersq     -0.001285577 2995
married     -0.019429116 2995
```

Report the coefficient on educ. Does it match your Wald estimate from Section 3.2?

The 2SLS regression yields a coefficient of 0.124893 on educ. This is essentially the same as the Wald estimate from Section 3.2.

4.3. Side-by-side comparison

Display both the OLS and 2SLS estimates in a single regression table using modelsummary(). Keep only the rows for educ and the model statistics (N, R^2).

```r
modelsummary(list("OLS" = ols, "2SLS" = iv),
coef_map = c("educ" = "Years of Education"),
gof_map = c("nobs", "r.squared"))
```

	OLS	2SLS
Years of Education	0.071	0.125
	(0.004)	(0.049)
Num.Obs.	3003	3003
R2	0.313	0.259

4.4. Why do OLS and 2SLS differ?

The OLS and 2SLS estimates of the return to education are likely to differ. Answer the following:

(a) Which is larger — OLS or 2SLS?

The 2SLS estimate of the return to education is larger.

(b) Card (1995) interprets the finding that 2SLS > OLS as evidence of downward bias in OLS. What could cause OLS to underestimate the return to education? (Hint: think about measurement error in educ, or about who the compliers are.)

OLS might underestimate the return to education because OLS takes into account the relationship between education level and weekly wages for all sampled individuals, whereas 2SLS captures the relationship between education level and weekly wages for those who actually attend more years of schooling because of their proximity to a 4-year college (those who are "compliers"). Compliers in this context are largely those who are less socioeconomically advantaged, so they benefit more from a marginal increase in education. These individuals "comply" with the treatment of education by taking advantage of a nearby 4-year college, so the LATE value from 2SLS is higher than that of ATE from OLS, which captures the relationship between education and wages for the general population.

Another potential cause of OLS underestimating the return to education is the issue of attenuation bias, which is caused by errors in the measurement of the independent variable (educ). With random noise and variability in the independent variable (error in educ), the coefficient from OLS is biased toward zero as the regression is fit to a noisier set of data points.

(c) Which estimate is more credible as a causal estimate of the effect of education on wages? Explain briefly.

The 2SLS estimate is more credible as a causal estimate because it uses the instrument of proximity to a 4-year college to isolate the effect of education level on wages. Whereas OLS has the problem of endogenous bias in the natural "assignment" of the treatment variable (in this case education level), 2SLS introduces an exogenous instrument (here it is nearc4) to control for any confounding from unobserved variables that might be correlated with the

treatment. By controlling for these variables as well as observed ones, the instrument and corresponding 2SLS method enables us to make a causal estimate.

4.5. Adding IQ as a control

The dataset includes IQ scores for a subset of respondents.

(a) Run OLS of lwage on educ, IQ, and all other controls. How does including IQ change the OLS coefficient on educ?

```
ols_IQ <- lm_robust(lwage ~ educ + IQ + black + smsa + south +
exper + expersq + married,
data = card, se_type = "HC2")

ols_IQ
```

```
               Estimate   Std. Error   t value      Pr(>|t|)     CI Lower
(Intercept)  4.624820266 0.1106111619 41.811515 7.250324e-277  4.407898361
educ         0.067568681 0.0050672360 13.334426  5.821735e-39  0.057631216
IQ           0.002500262 0.0007491806  3.337329  8.609879e-04  0.001031028
black       -0.120030879 0.0270219343 -4.441980  9.387896e-06 -0.173024170
smsa         0.162782345 0.0184132040  8.840522  2.006318e-18  0.126671818
south       -0.083180925 0.0183563649 -4.531449  6.195149e-06 -0.119179983
exper        0.083230844 0.0092631008  8.985203  5.709939e-19  0.065064780
expersq     -0.002350153 0.0004642726 -5.062012  4.518801e-07 -0.003260648
married     -0.029240623 0.0044083448 -6.633016  4.194221e-11 -0.037885922
               CI Upper   DF
(Intercept)  4.841742172 2051
educ         0.077506145 2051
IQ           0.003969496 2051
black       -0.067037588 2051
smsa         0.198892871 2051
south       -0.047181867 2051
exper        0.101396909 2051
expersq     -0.001439659 2051
married     -0.020595324 2051
```

After adding IQ as a covariate, the coefficient on educ decreased from 0.0715 without IQ to 0.0676 with IQ as a covariate. This is a decrease of 0.0039 on the coefficient for educ. IQ absorbed some of the ability bias (bias from the higher ability of participants with higher education), so the coefficient on educ was brought down.

(b) Now run 2SLS including IQ. Does the 2SLS estimate change substantially when IQ is added?

```
iv_IQ <- iv_robust(lwage ~ educ + IQ + black + smsa + south +
exper + expersq + married |
nearc4 + IQ + black + smsa + south +
exper + expersq + married,
data = card, se_type = "HC2")

iv_IQ
```

```
                 Estimate   Std. Error    t value     Pr(>|t|)     CI Lower
(Intercept)  4.1254833454 0.8039381042  5.1315932 3.143855e-07  2.548863208
educ         0.1093899302 0.0669607951  1.6336414 1.024876e-01 -0.021928311
IQ           0.0004228523 0.0034135214  0.1238757 9.014258e-01 -0.006271477
black       -0.1233784512 0.0280423967 -4.3997114 1.139525e-05 -0.178372993
smsa         0.1536399386 0.0239013796  6.4280783 1.602608e-10  0.106766434
south       -0.0857686314 0.0190205986 -4.5092498 6.872948e-06 -0.123070332
exper        0.1035065280 0.0339813212  3.0459830 2.348745e-03  0.036865035
expersq     -0.0027889965 0.0008630306 -3.2316312 1.250362e-03 -0.004481504
married     -0.0274086703 0.0053001348 -5.1713157 2.550570e-07 -0.037802878
                CI Upper   DF
(Intercept)  5.702103483 2051
educ         0.240708171 2051
IQ           0.007117182 2051
black       -0.068383910 2051
smsa         0.200513443 2051
south       -0.048466930 2051
exper        0.170148021 2051
expersq     -0.001096489 2051
married     -0.017014463 2051
```

Adding IQ to 2SLS, we get an educ coefficient of 0.1094, which is somewhat lower than the coefficient of 0.1249 from 2SLS without IQ as a covariate, but does not change substantially. This decrease of 0.0155 is larger than the decrease that occurred with the OLS coefficient when we added IQ. Adding IQ lowers the coefficient on educ in both cases.

(c) What does the comparison between part (a) and part (b) suggest about the source of the discrepancy between OLS and IV?

By adding IQ as a covariate, the discrepancy between OLS and 2SLS has decreased, but still remains. When we add IQ as a variable, 2SLS does not change substantially, meaning that it

is not as influenced by ability bias. 2SLS identifies a genuine causal effect, not a relationship influenced by ability bias. In this way, 2SLS is better than OLS because it is a stronger causal mechanism. Whereas OLS is driven by ability bias, 2SLS is not driven by ability bias, so it is a better causal estimate.

---

5. Interpretation and LATE

5.1. Who are the compliers?

In a randomized experiment, the ATE averages over everyone. But IV estimates a Local Average Treatment Effect (LATE) — the average effect for compliers: units whose treatment status changes when the instrument changes.

In this context, who are the compliers? Describe them in concrete terms: what type of person's educational attainment was increased because they happened to grow up near a 4-year college?

The compliers are those who attained a higher level of education (went to additional years of school) because they grew up near a four-year college. These are most likely people who attended at least some college (and thus gained more years of education) because they grew up near a college, as opposed to just graduating from high school and stopping at 12 years of education. More specifically, these compliers are people who would not have attended college if they had not lived near a 4-year college. They might be students from disadvantaged backgrounds who gain scholarship and in-state/county tuition from living nearby, or who learn about the benefits of college through recruiting and other exposure to higher education. The compliers are not people who would have attended college either way (always-takers), and they are not those who complete no further education even though they live near a college (never-takers).

5.2. Write the LATE interpretation

Write a single careful sentence interpreting the 2SLS coefficient as a LATE. Your sentence should specify: (i) The causal effect being measured (ii) For which specific group of people (iii) What source of variation identifies it

For men who attained a higher level of education (treatment) because they grew up near a four-year college (IV that randomizes the treatment), an additional year of education causes a 12.5% increase in weekly wages (outcome).

5.3 External validity

Card finds a 2SLS estimate around 12–13% return to each additional year of education. Is this LATE likely to be the same as the ATE (the return for the average person in the sample)?

Think about whether the compliers — men who went to more school because they happened to live near a college — are likely to have higher or lower returns to education than the typical person. Explain your reasoning.

No — the local average treatment effect (LATE) of education level on weekly wages is likely to be higher than the general average treatment effect (ATE). This is because those who are compliers (attain a higher level of education because they live near a college) will likely be those from more disadvantaged backgrounds, for whom returns to additional education are higher. Because these men normally would not have gone to college, the return on education in terms of weekly wages is higher for these individuals. As such, the LATE is higher than ATE given the characteristics of the compliers.